Fine-Tuned and Controller-Guided LLM Agent for WebShop

Group 34 · CS40008.01 Final Project

Abstract. We study language-model agents for WebShop, an interactive e-commerce environment requiring natural-language instruction understanding, search query reformulation, product comparison, and purchase decisions. Starting from a ReAct-style Qwen2.5-7B-Instruct baseline, we evaluate prompt-level query control, supervised fine-tuning (SFT), failure diagnosis, and a lightweight anti-loop strategy controller. Raw ReAct reaches 10% strong success on a 50-goal full synthetic evaluation. A keyword-only prompt improves the baseline to 20%. Naive SFT alone underperforms, mainly because the model repeatedly searches and fails to buy. After adding a conservative anti-loop controller, the unified SFT model reaches 48% strong success, the best result in our experiments.

1 Introduction

Web shopping is a realistic language-agent task: an agent must interpret user constraints, issue search queries, inspect product pages, select options, and decide when to buy. This project investigates whether a small, reproducible LLM agent can improve over a ReAct baseline in WebShop.

Our original plan followed a three-stage architecture: ReAct baseline, SFT, and reward-aware strategy control. During experimentation we found that naive SFT alone did not improve performance. Failure diagnosis showed that the SFT model often entered repeated-search/no-buy loops. This motivated a deterministic anti-loop controller as a lightweight strategy layer.

2 Task and Dataset

We use WebShop full mode with 1.18M products and a full Lucene search index. The agent receives a shopping instruction and interacts with a text-mode environment using actions such as `search[keywords]`, `click[ASIN]`, and `click[Buy Now]`.

The main evaluation uses synthetic goals 0-49, with at most 7 steps per episode. We report average reward and strong success rate, where strong success is defined as reward greater than 0.5.

3 Methods

Raw ReAct Agent

The raw baseline uses Qwen2.5-7B-Instruct with a ReAct prompt. At each step the model outputs a thought and a single WebShop action.

Keyword Prompt

Early failures showed that verbose search queries often hurt retrieval. We introduced a keyword-only prompt that asks the model to use short search queries with 3-6 important words.

Supervised Fine-Tuning

We trained LoRA adapters from WebShop-style trajectories. The unified SFT data combines cleaned trajectories and query examples. The goal was to teach the model better search and action patterns.

Anti-Loop Controller

Diagnosis showed that the SFT model often repeated the same search and failed to transition to click or buy. We implemented a deterministic controller with three main rules: repeated search triggers a click on a visible product; too many searches triggers a click; and conservative forced buying is applied near the step limit when already on a product page. This controller is a rule-based strategy-controller prototype, not a learned Bandit.

DPO Preparation

We identified DPO as a natural next stage, especially for action-level preferences such as repeated search versus click, or no-buy loop versus buy. DPO was not used as the main reported performance result.

4 Experiments and Results

System	Strong Success	Avg Reward	Notes
Raw ReAct	10%	0.077	Original prompt
Keyword baseline	20%	0.209	Short search prompt
Unified SFT	2%	0.012	Repeated search/no-buy
Unified SFT + conservative controller	48%	0.419	Best result

Controller Variant	Strong Success	Avg Reward
No controller	2%	0.012
Repeat-to-click only	22%	0.182
No forced buy	22%	0.192
Conservative forced buy	48%	0.419
Aggressive controller	48%	0.446

5 Analysis

The keyword prompt improves raw ReAct by shortening first queries. However, SFT alone fails: although it can generate more specific queries, it loses transition control and often searches repeatedly. The controller fixes this by forcing progress from search to click and from product inspection to purchase. This supports a two-layer design: a language model for textual understanding and a strategy controller for interaction policy.

6 Limitations and Future Work

The main evaluation uses only 50 synthetic goals, so the results should be validated on a larger split. The controller is deterministic rather than learned, so it should be interpreted as a prototype, not a completed Bandit algorithm. DPO was prepared as a next-stage method but not shown to improve performance in this report. Future work should construct aligned query-level and action-level preference pairs and replace the deterministic controller with a learned contextual Bandit.

7 Conclusion

Naive SFT alone does not solve WebShop. In our experiments, it weakens multi-step transition control and causes repeated-search/no-buy loops. However, when paired with a conservative anti-loop controller, the SFT agent reaches 48% strong success on a 50-goal full synthetic evaluation.

References

[1] Yao et al. *WebShop: Towards Scalable Real-World Web Interaction with Grounded Language Agents.* 2022.

[2] Yao et al. *ReAct: Synergizing Reasoning and Acting in Language Models.* 2022.

[3] Rafailov et al. *Direct Preference Optimization: Your Language Model is Secretly a Reward Model.* 2023.